

F200 / FLUID FILTER

Titanium housing · stainless media pack

PROJECT	Fluid filtration
DOCUMENT	F200-COMP-001
AUTHOR	Engineering concept
CHECKED	Review required
REV	A
DATE	2026-09-21

1 Overview

DESIGN INTENT

Two identical titanium cups capture a stainless five-layer filter pack. Fluid properties, filtration rating and all sweep inputs are editable.

- CP Ti Grade 2 housings; 316L frames; 316 stainless supports and Dutch-twill cloth.
- One closure weld. Integral inlet and outlet stubs share the same OD and wall.
- Calibration remains OPEN. Reference-screen and loading curves are sensitivities.

2 Requirements

REQUIREMENTS

ID	Requirement	Target
F-01	Operate at MEOP without leakage.	650 psi
F-02	Withstand proof pressure without leakage or unacceptable permanent deformation.	1300 psi
F-03	Withstand the specified minimum burst pressure.	2600 psi
F-04	Meet the selected filtration rating and particle retention efficiency.	5 µm; 99.9 % at 3 µm
F-05	Meet clean pressure-drop limit at nominal flow; characterize surge flow.	5 psi at 0.14 lb/s; surge 0.47 lb/s
F-06	Retain specified dirt mass within loaded pressure-drop limit at nominal flow.	0.318 g; 18 psi at 0.14 lb/s
F-07	Meet dry assembly mass limit.	1 lb
F-08	Approve compatibility and cleanliness for the selected fluid and operating state.	Selected fluid and state
F-09	Demonstrate media and support integrity at blocked-screen differential.	OPEN
F-10	Use the seven-component architecture.	7 parts

REFERENCES

- [1] Convair Division, Low Gravity Propellant Control Using Capillary Devices in Large Scale Cryogenic Vehicles: Related IRAD Studies, General Dynamics / NASA, 1970, GDC-DDB70-009, Eq. 2-2, Table 2-2, Sec. 2.1.7 — Reference Dutch-twill coefficients. Not a calibration for the selected filtration rating. ntrs.nasa.gov/api/citations/19710001290/download...
- [2] ATI, ATI CP Grade 2 titanium, ATI, Minimum tensile properties — Yield 275 MPa; ultimate 345 MPa. Confirm procurement form, temperature and weld allowables. www.atimaterials.com/Products/Pages/ati-cp2.aspx
- [3] IAPWS, IAPWS formulation for water viscosity, IAPWS, 2008, Default fluid example — Ambient-pressure water properties carried over from F100. Properties at operating pressure require confirmation. iapws.org/relguide/visc.pdf
- [4] Project engineering, Filter design inputs and pressure-drop method, Project record, 2026, F100 baseline and supplied September 2026 method — Pressure, flow and material concept inherited. Inlet OD and wall changed by user. Geometry and mesh packaging remain preliminary.

3 Inputs

OPERATING POINT

Input	Value	Workbook key
Inlet / outlet OD	0.25 in	stub_od
Wall	0.025 in	stub_wall
Filtration rating	5 µm	rating
MEOP	650 psi	p_meop
Proof	1300 psi	p_proof
Burst	2600 psi	p_burst
Nominal flow	0.14 lb/s	mdot_n
Surge flow	0.47 lb/s	mdot_s
Density	997.99 kg/m ³	rho
Dynamic viscosity	0.000979 Pa·s	mu
Temperature	70 °F	temperature_F

FLUID STATE

Selected fluid: Water (reference example) (liquid). Ambient-pressure properties at 70 F inherited from F100; operating-pressure correction OPEN. Use density and viscosity at the selected temperature and pressure. This is a single-phase, constant-property model; significant gas density change, two-phase flow and non-Newtonian behavior need a different flow model. Fluid / material compatibility remains OPEN.

4 Preliminary Analysis

BORE AND EXPOSED FACE

$$d_i = d_o - 2t, A_{\text{face}} = \pi \frac{d_{\text{face}}^2}{4}, \text{ and } U = \frac{\dot{m}}{\rho A_{\text{face}}}.$$

PRELIMINARY RESULTS

Quantity	Value
Derived bore	0.2 in
Exposed face area	2642.079 mm ²
Nominal approach velocity	0.024 m/s
Surge approach velocity	0.081 m/s

INLET VELOCITY AND DYNAMIC PRESSURE

Flow	Bore velocity	Dynamic pressure
Nominal	3.139 m/s	0.713 psi
Surge	10.54 m/s	8.039 psi

HOUSING LOSS SENSITIVITY

Assumed K	Nominal loss	Surge loss
0	0 psi	0 psi
1	0.713 psi	8.039 psi
3	2.14 psi	24.118 psi

5 Sizing

PRESSURE BOUNDARY

The closed-end thick-cylinder screen uses $\sigma_{\text{VM}} = \sqrt{3p \frac{r_o^2}{r_o^2 - r_i^2}}$ at the bore. Compare proof stress with yield and burst stress with minimum tensile strength. These local elastic comparisons exclude the cone, transitions, weld and fatigue. They do not establish proof or burst qualification. Material minima: ^[2].

LOCAL PRESSURE SCREENS

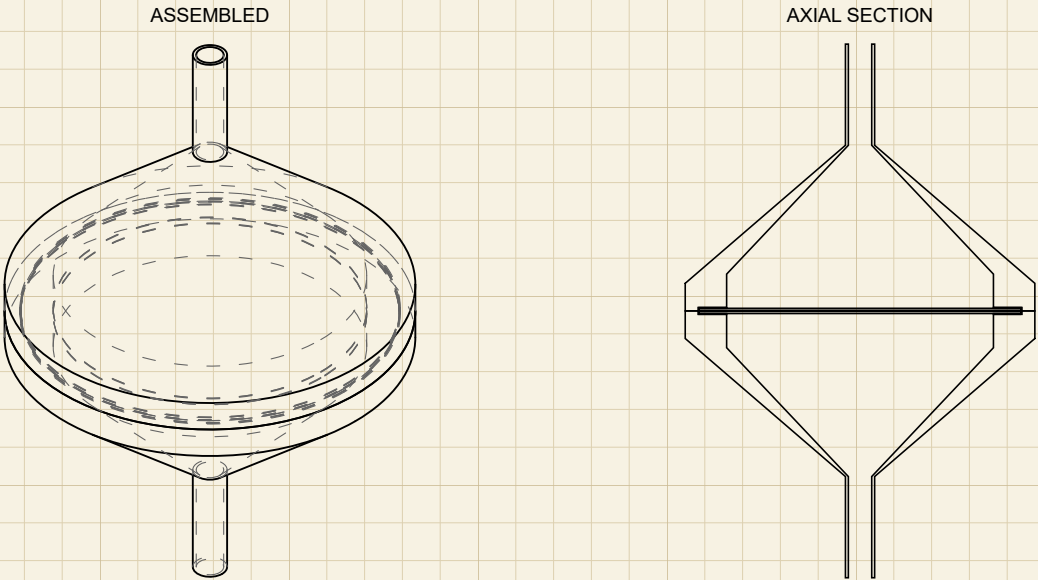
Location / case	VM stress	Strength / stress
Stub / Proof	43.124 MPa	6.377
Stub / Burst	86.248 MPa	4.000
Weld rim / Proof	109.376 MPa	2.514
Weld rim / Burst	218.751 MPa	1.577

MASS ESTIMATE

Quantity	Value
Estimated dry mass	0.594 lb
With growth allowance	0.713 lb
Mass limit	1 lb

6 Design

ASSEMBLY AND AXIAL SECTION



SEVEN COMPONENTS

Component	Qty	Material / definition
Turned cup	2	CP Ti Grade 2; identical halves with integral stubs
Annular frame	2	316L; raw 0.2 mm, assembled 0.17 mm
Coarse support	2	316 stainless, 40 x 40; assembled 0.48 mm
Fine cloth	1	316 stainless Dutch twill; 5 µm; thickness 0.18 mm provisional

INTERFACES

Exposed face diameter 58 mm; pack OD 70 mm; body OD 76 mm. Mass uses revolved housing volume, raw frame volume and provisional cloth areal weights. Coining, media thickness, edge sealing and blocked-screen support capacity remain OPEN.

7 Analysis

BLOCKED Dependent work is waiting on missing inputs.

CLEAN SCREEN MODEL

Use superficial velocity $U = \frac{\dot{m}}{\rho A_{\text{face}}}$ and $\Delta p_0 = C_v \mu U + C_i \rho U^2$. For the published cloths, $C_v = A_1 \frac{L}{D_a^2}$ and $C_i = A_2 \frac{L}{D_a}$. D_a is measured average capillary diameter, distinct from filtration rating. No additional porosity correction is applied. ^[1]

The reference curves represent individual clean screens. A finished-pack fit must include the two support layers and assembly effects. Housing loss is separate: $\Delta p_{\text{housing}} = K \rho \frac{V_{\text{bore}}^2}{2}$. Its coefficient is also uncalibrated.

PUBLISHED REFERENCE CLOTHS

Cloth	Da (µm)	L (mm)	A1	A2
200 x 1400	22.800	0.14732	190	18
165 x 1400	28.300	0.15240	150	16
200 x 600	36.700	0.13970	52	3

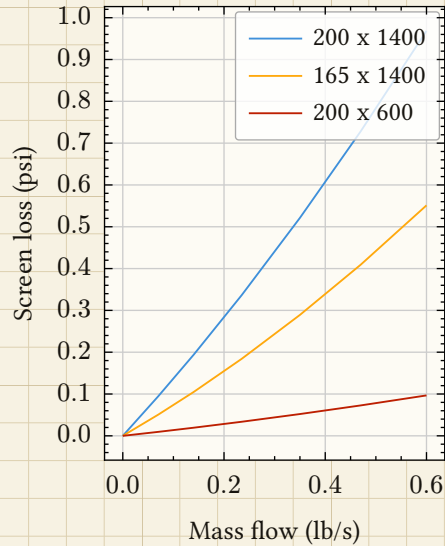
MICRON-RATING SWEEP

Rating	Nominal pack loss	Calibration
1 µm	OPEN	Cv / Ci and evidence missing
3 µm	OPEN	Cv / Ci and evidence missing
5 µm	OPEN	Cv / Ci and evidence missing
10 µm	OPEN	Cv / Ci and evidence missing
20 µm	OPEN	Cv / Ci and evidence missing

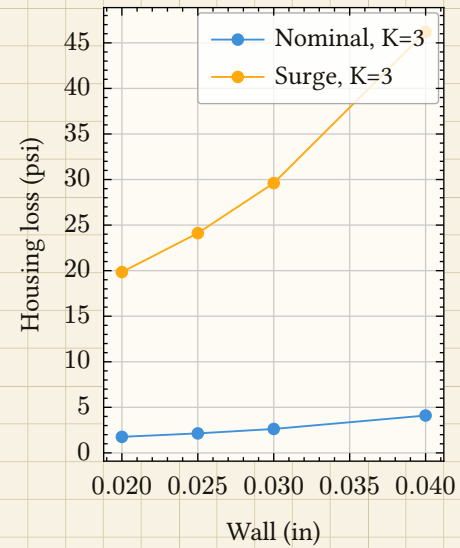
SELECTED FINISHED-PACK PREDICTION

BLOCKED Selected rating: finished-pack C_v , C_i and supporting evidence are OPEN. Reference sweeps remain available.

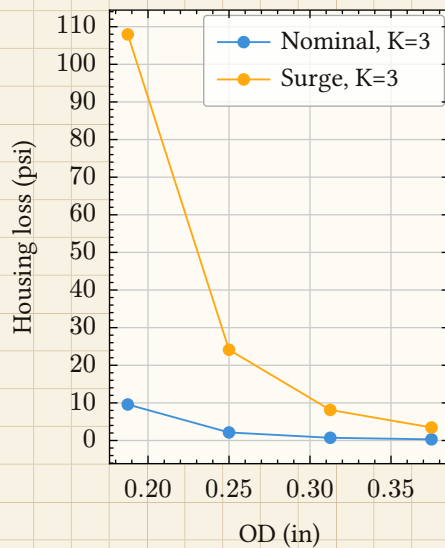
FLOW SENSITIVITY – REFERENCE SCREENS ONLY



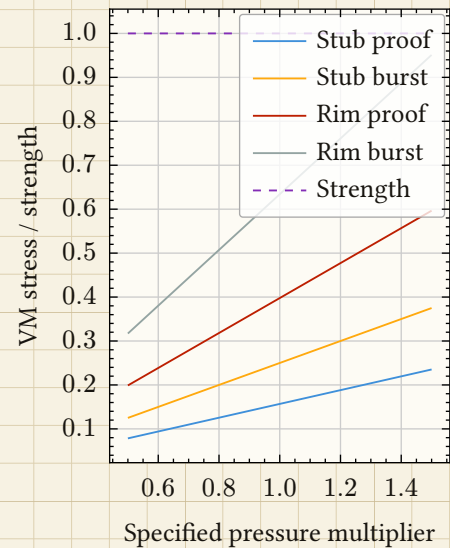
INLET WALL SENSITIVITY – ASSUMED K



INLET OD SENSITIVITY – ASSUMED K



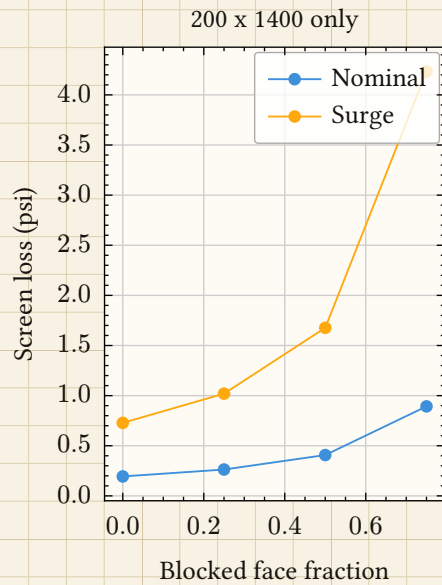
PROOF AND BURST PRESSURE SENSITIVITY



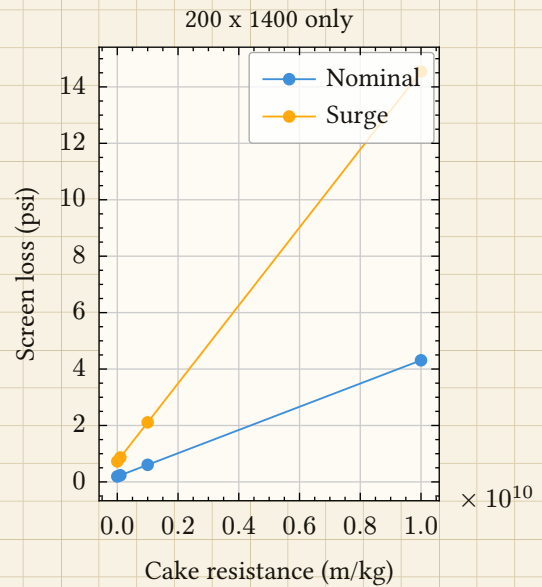
LOADING SENSITIVITIES

For blocked-area fraction b , let $F = 1 - b$ and use $\Delta p_b = C_v \mu \frac{U}{F} + C_i \rho \frac{U^2}{F^2}$. Alternatively, a cake adds $\mu U \alpha_c \frac{m_d}{A_{face}}$ to clean loss. The curves below apply each idealization separately to one reference cloth. Blockage is not inferred from dirt mass. Cake resistance, retention performance and media strength require test data. ^[4]

BLOCKED-AREA SENSITIVITY — REFERENCE CLOTH



CAKE SENSITIVITY — ASSUMED RESISTANCE



CALIBRATION AND LIMITS

Fit measured clean pack data to the two terms μU and ρU^2 at a known fluid state. Record both coefficients and evidence for each rating. Micron rating alone supplies no resistance curve. Fit housing loss separately or measure the complete assembly and identify that scope explicitly. The clean and loaded loss requirements apply at nominal flow. Surge curves show sensitivity; no separate surge loss limit has been specified.

8 Manufacturing

PROCESS

id	operation	acceptance
M1	Turn two identical CP Ti Grade 2 cups with integral stubs	Verify OD, wall, bore, lands and weld preparation against released drawing
M2	Cut 316L annular frames and 316 stainless support / Dutch-twill cloth	Confirm alloy, thickness, rating, areal mass and finished-pack flow curve
M3	Assemble frame / support / fine / support / frame stack	Inspect edge capture and bypass seal. Coining and pack compression require development
M4	Orbital weld the titanium closure seam	Qualify procedure, shielding, inspection and leak acceptance
M5	Clean and identify the assembly	Confirm cleanliness and material compatibility for the selected fluid

9 Test

OPEN Test results or evidence are not recorded for every row.

PLANNED VERIFICATION

id	requirement	procedure	criterion
T1	F-01	Leak test at MEOP	No leakage at specified MEOP
T2	F-02	Proof pressure test	No leakage or unacceptable permanent deformation at proof pressure
T3	F-03	Destructive burst test	Withstand minimum specified burst pressure
T4	F-04	Particle challenge / retention test	Confirm rating and efficiency at specified particle size
T5	F-05	Clean flow bench at controlled fluid state	Meet nominal clean loss limit; characterize through surge flow
T6	F-06	Dirt loading at nominal flow	Retain specified dirt mass within loaded loss limit
T7	F-07	Weigh final dry assembly	Meet dry mass limit
T8	F-08	Fluid compatibility and cleanliness review	Approve selected fluid, state and all wetted materials
T9	F-09	Blocked-screen differential test	Define required differential; verify support and media integrity

10 Compliance

OPEN 9 requirement(s) need verification.

COMPLIANCE

ID	Requirement	Assessment	Status
F-01	Operate at MEOP without leakage.	Awaiting test evidence.	OPEN
F-02	Withstand proof pressure without leakage or unacceptable permanent deformation.	Awaiting test evidence.	OPEN
F-03	Withstand the specified minimum burst pressure.	Awaiting test evidence.	OPEN
F-04	Meet the selected filtration rating and particle retention efficiency.	Awaiting test evidence.	OPEN
F-05	Meet clean pressure-drop limit at nominal flow; characterize surge flow.	Finished-pack and housing calibration OPEN.	OPEN
F-06	Retain specified dirt mass within loaded pressure-drop limit at nominal flow.	Loading and retention test evidence OPEN.	OPEN
F-07	Meet dry assembly mass limit.	CAD estimate available; finished mass and cloth areal weights OPEN.	OPEN
F-08	Approve compatibility and cleanliness for the selected fluid and operating state.	Awaiting inspection evidence.	OPEN
F-09	Demonstrate media and support integrity at blocked-screen differential.	Required differential and support capacity OPEN.	OPEN
F-10	Use the seven-component architecture.	7 == 7; Generated seven-solid CAD assembly	PASS